

7.2 mi³

The most plant available water the top 24 inches (2 feet) of Kansas soils can hold, with the whole state at field capacity.

≈ 7.9 trillion gallons · ≈ 24 million acre-feet

Burj Khalifa
2,717 ft

Empire State
1,454 ft

Eiffel Tower
1,083 ft

Great Pyramid
455 ft



Gathered into one marble, it would stand more than 4½ times taller than the world's tallest building.

12 million

Olympic swimming pools —
one pool = 660,000 gal

94%

of full Lake Mead, the largest man-made reservoir in the U.S.

5.6 in

of a single rainfall event soaking the entire state at once

1 mile · everything drawn to the same scale

CLOSER TO HOME

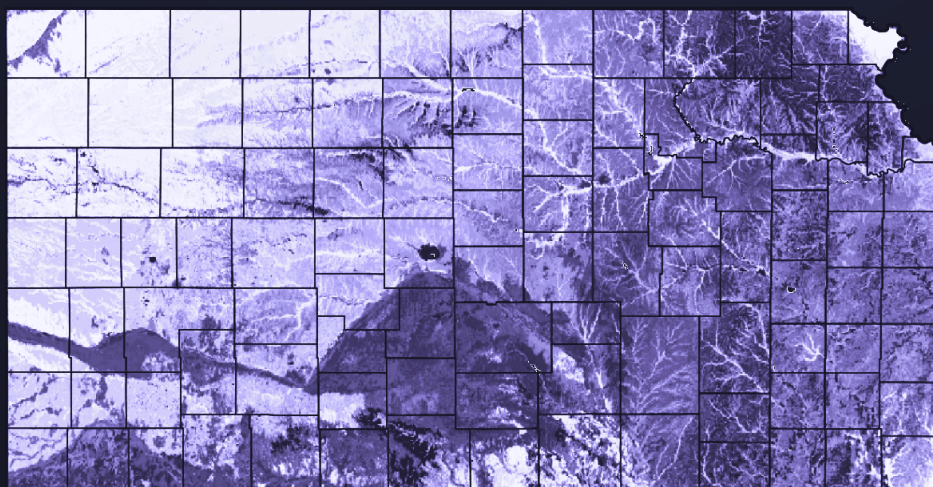
5½ times

That's how many times this water could fill every lake and pond in Kansas (≈ 4.4 million acre-ft).

Or, the ten largest lakes one at a time:

Millford	366,000 acre-ft	66×
Wilson	237,000 acre-ft	103×
Tuttle Creek	232,000 acre-ft	105×
Waconda	209,000 acre-ft	116×
Perry	193,000 acre-ft	126×
Cedar Bluff	169,000 acre-ft	144×
Cheney	156,000 acre-ft	155×
El Dorado	156,000 acre-ft	155×
Melvorn	148,000 acre-ft	164×
Clinton	116,000 acre-ft	210×

WHERE IT IS STORED



≤ 4.5 in | ≥ 6.5 in

Deep silt loams in the northwest store the most; sandy soils along the Arkansas River and in south-central Kansas store the least.

WHY IT MATTERS

Crops, pastures, and native grasslands draw on this stored water to sustain transpiration and produce grain and forage between rains. How full this reservoir is at any moment, known as the fraction of available water, is a robust indicator of agricultural drought.

Plant available water is what a soil holds after excess water drains away (field capacity) minus what it holds too tightly for roots to extract (permanent wilting point). We report the top 24 inches (2 feet), the most dynamic and densely rooted part of the soil profile, where water content increases within hours of rainfall and declines over the following days through soil evaporation and plant uptake. Both limits were estimated with custom pedotransfer functions developed by the K-State Soil Water Processes Lab, using clay and sand content maps from the USDA Natural Resources Conservation Service. Figures rounded: full Lake Mead ≈ 26 million acre-ft; Kansas land area ≈ 81,800 mi²; lakes and ponds at normal pool (Rahmani et al. 2018; USACE National Inventory of Dams; Callihan 2013). Created by Andres Patrignani, Soil Water Processes Lab, Agronomy, Kansas State University.